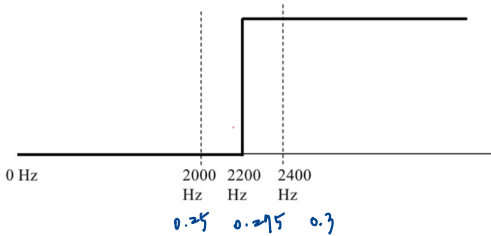


(1) Design a Mini-max **highpass** FIR filter such that

(40 scores)

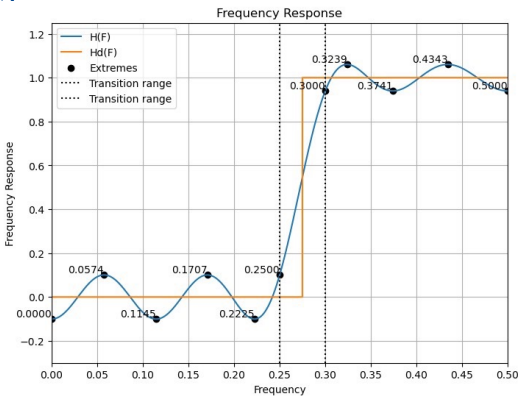
- ① Filter length = 19, ② Sampling frequency $f_s = 8000\text{Hz}$, ③ Stop Band 0~2000Hz
- ④ Transition band: 2000~2400 Hz,
- ⑤ Weighting function: $W(F) = 1$ for passband, $W(F) = 0.6$ for stop band . ⑥ Set $\Delta = 0.0001$ in Step 5.

Show (a) the frequency response, (b) the impulse response $h[n]$, and (c) the maximal error for each iteration.

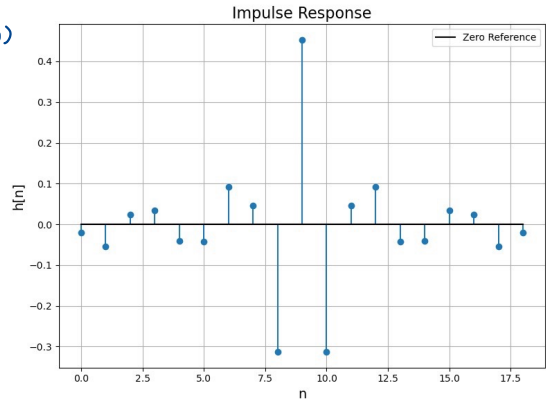


ideal filter : $H_d(F) = 0$ for $0 \leq F < 0.25$
 $H_d(F) = 1$ for $0.25 \leq F \leq 0.5$
 transition band : $0.25 < F < 0.3$
 weighting function : $W(F) = 0.6$ for $0 \leq F \leq 0.25$
 $W(F) = 1$ for $0.3 \leq F \leq 0.5$

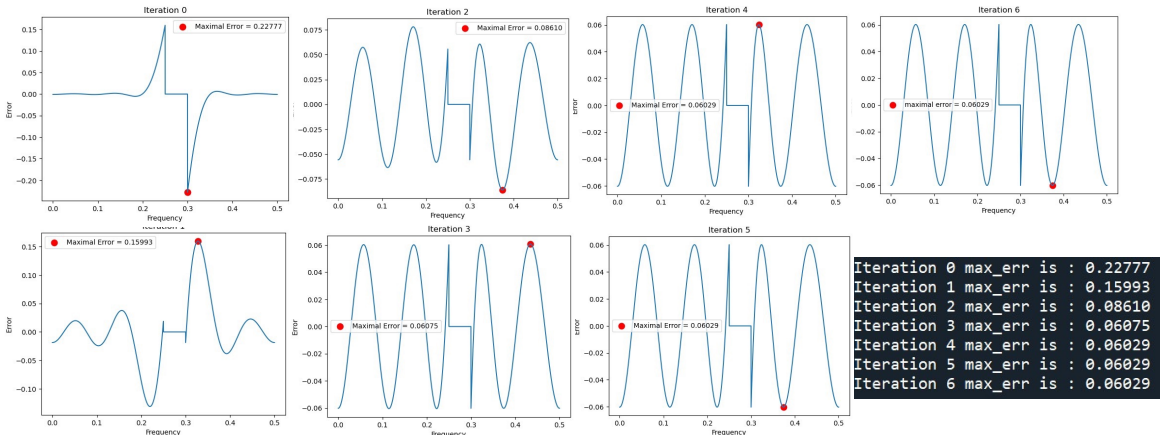
(a)



(b)



(c)



(2) (a) How do we convert convolution into an addition operation?

(b) What are two main advantages of the FT in engineering?

2 個

(c) From the view point of implementation, what are the disadvantages of the discrete Fourier transform?

(15 scores)

$$(a) \quad y[n] = x[n] * h[n]$$

↓ Fourier Transform

$$Y(f) = X(f) H(f)$$

↓ log

$$\log(Y(f)) = \log(X(f)) + \log(H(f))$$

(b) · spectrum analysis

· convolution → multiplication

$$y(t) = x(t) * h(t) \rightarrow Y(f) = X(f) H(f)$$

$$= \int x(t-\tau) h(\tau) d\tau$$

$$(c) \quad DFT: X[n] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi mn}{N}}$$

① 無理数, ② 複数 在運算上會比較複雜.

(3) Discuss how to implement $y[n] = x[n] * h[n]$ efficiently where $h[n] = (0.7^n + (-0.6)^{n+1})u[n]$, $u[n]$: unit step function (10 scores)

$$\begin{aligned} y[n] &= x[n] * (0.7^n + (-0.6)^{n+1}) u[n] \\ &= x[n] * (0.7^n u[n] + (-0.6)^{n+1} u[n]) \\ Z \left(\begin{aligned} Y(z) &= X(z) H(z) \\ &= X(z) \cdot \left(\frac{1}{1-0.7z^{-1}} + \frac{-0.6}{1+0.6z^{-1}} \right) \\ &= X(z) \cdot \left(\frac{1 + 0.6z^{-1} - 0.6 + 0.42z^{-1}}{(1-0.7z^{-1})(1+0.6z^{-1})} \right) \end{aligned} \right. \end{aligned}$$

$$\begin{aligned} H(z) &= \sum_{n=0}^{\infty} h(n) z^{-n} \\ &= \sum_{n=0}^{\infty} 0.7^n z^{-n} + \sum_{n=0}^{\infty} (-0.6)^{n+1} z^{-n} \\ &= \sum_{n=0}^{\infty} (0.7z^{-1})^n + \sum_{n=0}^{\infty} (-0.6z^{-1})^{n+1} \\ &= \frac{1}{1-0.7z^{-1}} + \frac{0.6}{1+0.6z^{-1}} \end{aligned}$$

$$(1 - 0.1z^{-1} - 0.42z^{-2}) Y(z) = X(z) (0.4 + 1.02z^{-1})$$

$$y[n] - 0.1y[n-1] - 0.42y[n-2] = 0.4x[n] + 1.02x[n-1]$$

$$y[n] = 0.1y[n-1] + 0.42y[n-2] + 0.4x[n] + 1.02x[n-1] *$$

$$\frac{a(1-r^n)}{1-r}$$

$$-1 < r < 1 \Rightarrow \frac{a}{1-r}$$

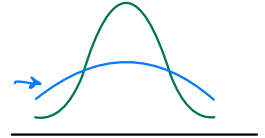
$$r \leq -1, r > 1 \Rightarrow \infty$$

(4) Why (a) the step invariance method and (b) the bilinear transform can reduce or avoid the aliasing effect in IIR filter design? (10 scores)

(a) 將 $h_n(t)$ 和 $u(t)$ 做 convolution ,
 $h_{a,n}(t) = h_n(t) * u(t) = \int_{-\infty}^{\infty} h_n(\tau) u(t-\tau) d\tau$ $\rightarrow u(t-\tau) = 1$ if $t-\tau \geq 0$
 $= \int_{-\infty}^t h_n(\tau) d\tau$ (對 $h_n(t)$ 做積分).

$$FT \rightarrow H_{a,n}(f) = \frac{H_n(f)}{j2\pi f}$$

f 較大, $H_{a,n}(f)$ 就可以被壓下來.
 f 較小, $H_{a,n}(f)$ 就會被放大. $\rightarrow$ aliasing effect 減少



(b) $f_{new} = \frac{f_s}{\pi} \arctan\left(\frac{2\pi}{c} f_{old}\right)$, $\rightarrow$ mapping function

用 mapping function 讓 f 原本是無限長 轉換成在 $-\frac{f_s}{2} \sim \frac{f_s}{2}$ 之間 $\rightarrow$ 防 aliasing effect

(5) Design the 7-point FIR filter in the MSE sense where the ideal filter is $H_d(F) = 1$ for $|F| < 0.25$,

$$H_d(F) = 0 \text{ for } 0.25 < |F| < 0.5$$

(10 scores)

$$H_d(F) = 1 \text{ for } |F| < 0.25$$

$$H_d(F) = 0 \text{ for } 0.25 < |F| < 0.5$$

$$S[0] = \int_{-\frac{1}{2}}^{\frac{1}{2}} H_d(F) dF = \int_{-\frac{1}{4}}^{\frac{1}{4}} 1 dF = 0.5$$

$$S[n] = 2 \int_{-\frac{1}{2}}^{\frac{1}{2}} \cos(2\pi n F) H_d(F) dF = 2 \int_{-\frac{1}{4}}^{\frac{1}{4}} \cos(2\pi n F) dF$$

$$= 2 \times \frac{1}{2\pi n} \left(\sin(2\pi n F) \right) \Big|_{-\frac{1}{4}}^{\frac{1}{4}}$$

$$= \frac{1}{\pi n} \left(\sin\left(\frac{\pi n}{2}\right) + \sin\left(\frac{\pi n}{2}\right) \right)$$

$$= \frac{1}{\pi n} \sin\left(\frac{\pi n}{2}\right) \rightarrow S[1] = 0.63662$$

$$S[2] \approx 0$$

$$S[3] \approx -0.21221$$



$h[0]$	$h[1]$	$h[2]$	$h[3]$	$h[4]$	$h[5]$	$h[6]$
"	"	"	"	"	"	"
$\frac{S[3]}{2}$	$\frac{S[2]}{2}$	$\frac{S[1]}{2}$	$\frac{S[0]}{2}$	$\frac{S[1]}{2}$	$\frac{S[2]}{2}$	$\frac{S[3]}{2}$
"	"	"	"	"	"	"
-0.106105	0	0.31831	0.5	0.31831	0	-0.106105

*

(6) (a) Write two reasons why the transition band plays a critical role in Minimax filter design.

$\Delta f = ?$

(b) Estimate the pass and stop band ripples if filter length = 21, $\Delta_f = 0.0002$, and the transition band is 1950~2050 Hz.

(c) Estimate the pass and stop band ripples if filter length = 31, $\Delta_f = 0.0001$, and the transition band is 1750~2250 Hz.

In (b)(c), suppose that pass and stop band ripples are equal.

(15 scores)

(A) ① the error in the transition band is ignored.

如果每個 Δf 的誤差都要考慮的話 最大誤差不可能小於 0.5

② 減少誤差, transition band 的寬度會影響誤差的大小

(b) $N = 21 \quad \frac{2}{3} \frac{1}{\Delta f} \log_{10} \left(\frac{1}{10 \delta_1 \delta_2} \right)$

$$\Delta f = (2050 - 1950) \times 0.0002 \\ = 0.02$$

$$\delta_1 \delta_2 = 10^{-\frac{2}{3} N \Delta f - 1}$$

$$\delta_1 \delta_2 = 10^{-\frac{2}{3} \times 21 \times 0.02 - 1}$$

$$\delta_1 = \delta_2 = 10^{(-\frac{2}{3} \times 21 \times 0.02 - 1) \frac{1}{2}} = 0.153$$

(c) $\Delta f = (2250 - 1750) \times 0.0001$
 $= 0.05$

$$\delta_1 \delta_2 = 10^{-\frac{2}{3} \times 31 \times 0.05 - 1}$$

$$\delta_1 = \delta_2 = 10^{(-\frac{2}{3} \times 31 \times 0.05 - 1) \frac{1}{2}} = 0.022$$

(7) differentiation 和 integration 各是 low pass, high pass ?

differentiation $\rightarrow H(f) = j2\pi f$

頻率響應會因 f 越大而放大, 所以是 high pass

integration $\rightarrow H(f) = \frac{1}{j2\pi f}$

頻率響應會因 f 越大而衰減, 所以是 low pass.